

CMD Troubleshooting Commands Guide

Essential CMD Commands for Troubleshooting

System Information & Diagnostics

Command	Description	Example
systeminfo	Detailed system configuration	<code>systeminfo findstr "OS Name"</code>
hostname	Display computer name	<code>hostname</code>
whoami	Current user	<code>whoami</code>
ver	Windows version	<code>ver</code>
wmic os get	OS information via WMI	<code>wmic os get caption, version, lastbootuptime</code>
wmic cpu get	CPU information	<code>wmic cpu get name, maxclockspeed, numberofcores</code>
wmic memorychip get	RAM information	<code>wmic memorychip get capacity, speed, manufacturer</code>
wmic diskdrive get	Disk information	<code>wmic diskdrive get model, size, interfacetype</code>
date	Display/set date	<code>date /T</code>
time	Display/set time	<code>time /T</code>
set	Environment variables	<code>set findstr PATH</code>
echo %VAR%	Display variable value	<code>echo %COMPUTERNAME%</code>
tasklist	Running processes	<code>tasklist /v findstr chrome</code>
tasklist /svc	Processes with services	<code>tasklist /svc findstr svchost</code>
driverquery	Installed drivers	<code>driverquery /v</code>
driverquery /si	Signed drivers	<code>driverquery /si</code>

Network Troubleshooting

Command	Description	Example
ping	ICMP echo test	<code>ping google.com -n 4</code>
tracert	Route trace	<code>tracert -d google.com (no DNS)</code>
pathping	Path with packet loss stats	<code>pathping google.com</code>
nslookup	DNS query	<code>nslookup google.com</code>
ipconfig	Network configuration	<code>ipconfig /all</code>
ipconfig /release	Release IP	<code>ipconfig /release</code>
ipconfig /renew	Renew IP	<code>ipconfig /renew</code>
ipconfig /flushdns	Clear DNS cache	<code>ipconfig /flushdns</code>

Command	Description	Example
<code>ipconfig /displaydns</code>	Show DNS cache	<code>ipconfig /displaydns</code>
<code>netstat</code>	Network connections	<code>netstat -an</code>
<code>netstat -b</code>	Connections with executables	<code>netstat -b -n</code>
<code>netstat -o</code>	Connections with process IDs	<code>netstat -ano findstr :80</code>
<code>route print</code>	Routing table	<code>route print -4</code>
<code>arp -a</code>	ARP cache	<code>arp -a</code>
<code>netsh interface show</code>	Network interfaces	<code>netsh interface show interface</code>
<code>netsh wlan show</code>	WiFi profiles	<code>netsh wlan show profiles</code>
<code>netsh wlan show networks</code>	Available networks	<code>netsh wlan show networks mode=bssid</code>
<code>net share</code>	Shared resources	<code>net share</code>
<code>net use</code>	Mapped drives	<code>net use</code>
<code>telnet</code>	Port connectivity test	<code>telnet google.com 80</code>
<code>curl</code>	HTTP requests	<code>curl https://example.com -v</code>
<code>tracertcp</code>	TCP traceroute	<code>tracertcp google.com:443</code>

Disk & File System

Command	Description	Example
<code>chkdsk</code>	Check disk errors	<code>chkdsk C: /f /r</code>
<code>diskpart</code>	Disk management	<code>diskpart</code> (list disk, select disk, detail)
<code>wmic logicaldisk get</code>	Logical disk info	<code>wmic logicaldisk get caption, size, freespace, description</code>
<code>fsutil volume diskfree</code>	Disk free space	<code>fsutil volume diskfree C:</code>
<code>fsutil fsinfo drives</code>	Available drives	<code>fsutil fsinfo drives</code>
<code>dir</code>	List directory contents	<code>dir /s /b *.log</code>
<code>dir /a</code>	List all including hidden	<code>dir /a /s C:*.exe</code>
<code>tree</code>	Directory structure	<code>tree /f C:\Projects</code>
<code>copy</code>	Copy files	<code>copy source.txt dest.txt</code>
<code>xcopy</code>	Advanced file copy	<code>xcopy C:\Source D:\Dest /E /I</code>
<code>robocopy</code>	Robust file copy	<code>robocopy C:\Source D:\Dest /MIR /R:3 /W:10</code>
<code>move</code>	Move files	<code>move file.txt C:\NewLocation\</code>
<code>del</code>	Delete files	<code>del *.tmp /s</code>

Command	Description	Example
rmdir	Delete directory	rmdir /s /q C:\Temp
mkdir	Create directory	mkdir C:\NewFolder
attrib	View/change attributes	attrib +h +s folder
assoc	File associations	assoc .txt
ftype	File type mapping	ftype txtfile
takeown	Take ownership	takeown /F C:\Folder /R
icacls	View/change permissions	icacls C:\Folder /grant username:F

Process Management

Command	Description	Example
tasklist	List processes	tasklist /FI "memusage gt 50000"
tasklist /M	Processes with loaded DLLs	tasklist /M user32.dll
taskkill /PID	Kill process by ID	taskkill /PID 1234 /F
taskkill /IM	Kill process by name	taskkill /IM chrome.exe /F
wmic process	Process management	wmic process where name="explorer.exe" get processid,parentprocessid
start	Start program/process	start notepad.exe
timeout	Wait/ pause execution	timeout /t 30 /nobreak

Services Management

Command	Description	Example
sc query	List services	sc query type= service state= all
sc query state=	Filter services	sc query state= inactive
sc queryex	Extended service info	sc queryex wuauserv
sc start	Start service	sc start wuauserv
sc stop	Stop service	sc stop wuauserv
sc config	Configure service	sc config wuauserv start= auto
sc failure	Service recovery options	sc failure wuauserv reset= 86400 actions= restart/60000
net start	List running services	net start
net stop	Stop service	net stop "Windows Update"

Command	Description	Example
wmic service	Service management via WMI	wmic service where name="wuauserv" get displayname,startmode,state

Event Logs & Troubleshooting

Command	Description	Example
wevtutil	Event log utility	wevtutil qe System /c:10 /rd:true /f:text
wevtutil query-events	Query events	wevtutil qe Application /c:50 /f:xml /rd:true
wevtutil el	List event logs	wevtutil el
eventvwr	Open Event Viewer	eventvwr.msc
findstr	Search in files/output	systeminfo findstr "Boot Time"
find	Search in files	find "error" C:\Logs\app.log
type	Display file content	type C:\Windows\WindowsUpdate.log
more	Paginated output	type file.txt more

User & Group Management

Command	Description	Example
net user	List users	net user
net user username	User details	net user Administrator
net user /add	Create user	net user NewUser Password123 /add /active:yes
net user /delete	Delete user	net user NewUser /delete
net localgroup	List groups	net localgroup
net localgroup groupname	Group members	net localgroup Administrators
net localgroup /add	Add group	net localgroup NewGroup /add
net localgroup /add /delete	Add/remove member	net localgroup Administrators User /add
wmic useraccount	User accounts via WMI	wmic useraccount where name='username' get sid
whoami /groups	Current user groups	whoami /groups
whoami /all	All user info	whoami /all

System Boot & Recovery

Command	Description	Example
bootrec /fixmbr	Fix MBR	bootrec /fixmbr
bootrec /fixboot	Fix boot sector	bootrec /fixboot
bootrec /rebuildbcd	Rebuild BCD	bootrec /rebuildbcd
sfc /scannow	System File Checker	sfc /scannow
sfc /verifyfile	Verify specific file	sfc /verifyfile=C:\Windows\System32\kernel32.dll
DISM /Online	Deployment Imaging	DISM /Online /Cleanup-Image /RestoreHealth
DISM /Source	With source files	DISM /Online /Cleanup-Image /RestoreHealth /Source:D:\Sources\install.wim
bcdedit	Boot configuration	bcdedit /enum
bcdedit /set	Modify boot settings	bcdedit /set {default} bootmenupolicy legacy
chkdsk /f	Check disk (fix)	chkdsk C: /f /r /x
chkdsk /scan	Online scan	chkdsk /scan

Performance & Troubleshooting

Command	Description	Example
perfmon	Performance Monitor	perfmon.msc
typeperf	Performance counter	typeperf "\Processor(_Total)\% Processor Time" -sc 10
wmic pagefile	Page file info	wmic pagefile list
wmic computersystem	System info	wmic computersystem get totalphysicalmemory
getmac	MAC address	getmac /v
systeminfo /s	Remote system info	systeminfo /s SERVER01 /u USERNAME
systeminfo /fo csv	Format output	systeminfo /fo csv

PowerShell vs CMD Comparison

CMD Command	PowerShell Equivalent	Description
ipconfig	Get-NetIPConfiguration	Network config

CMD Command	PowerShell Equivalent	Description
ping	Test-Connection	Network test
netstat	Get-NetTCPConnection	TCP connections
tasklist	Get-Process	Process list
taskkill	Stop-Process	Kill process
sc query	Get-Service	Service list
net start	Start-Service	Start service
findstr	Select-String	Text search
wmic	Get-CimInstance	WMI queries
systeminfo	Get-ComputerInfo	System info

Common Troubleshooting Scenarios

Case: Network Connectivity Issues

```
cmd
:: Test network stack
ping 127.0.0.1
ping %COMPUTERNAME%
ping gateway_ip
ping 8.8.8.8

:: Check DNS
nslookup google.com
ipconfig /displaydns
ipconfig /flushdns

:: Check network config
ipconfig /all
netstat -ano

:: Reset TCP/IP
netsh int ip reset
netsh winsock reset
```

Case: Service Failed to Start

```
cmd
:: Check service
sc query service_name
sc queryex service_name

:: Check dependencies
sc enumdeps service_name

:: Check service configuration
sc qc service_name

:: View event logs
```

```
wevtutil qe System /c:20 /f:text /rd:true | findstr "service_name"
```

```
:: Force start with verbose  
sc start service_name
```

Case: System Performance Issues

cmd

```
:: Check CPU usage  
tasklist /v /FI "memusage gt 100000"  
  
:: Check memory  
wmic os get totalvisiblememorysize,freephysicalmemory  
wmic pagefile list  
  
:: Check disk I/O  
wmic diskdrive get status, model, size  
  
:: Check running processes by CPU  
tasklist /v /FI "cputime gt 00:10:00"  
  
:: Check event logs for errors  
wevtutil qe System /c:50 /f:text /rd:true | findstr /i "error critical"
```

Case: Disk Space Issues

cmd

```
:: Check free space  
wmic logicaldisk get caption, freespace, size, description  
fsutil volume diskfree C:  
  
:: Find large files  
dir /s /b /a-d C:\ | findstr /v "Windows" | for %f in (dir) do @echo %~zf %f  
  
:: Check disk for errors  
chkdsk C: /scan  
chkdsk C: /f /r  
  
:: Clean temporary files  
del /f /s /q C:\Temp\*.*  
del /f /s /q %TEMP%\*.*
```

Case: Windows Update Issues

cmd

```
:: Check Windows Update service  
sc query wuauserv  
sc query bits  
sc query trustedinstaller  
  
:: Reset Windows Update  
net stop wuauserv  
net stop bits  
ren C:\Windows\SoftwareDistribution SoftwareDistribution.old
```

```
net start bits
net start wuauserv

:: Check update history
wevtutil qe System /c:50 /f:text /rd:true | findstr "WindowsUpdate"

:: Run DISM
DISM /Online /Cleanup-Image /RestoreHealth

:: Run SFC
sfc /scannow
```

Case: User Access/Permissions Issues

cmd

```
:: Check user info
whoami /all
whoami /groups
net user %USERNAME%

:: Check group membership
net localgroup Administrators
net localgroup Users

:: Check permissions
icacls C:\Folder\file.txt
icacls C:\Folder /t

:: Fix permissions (take ownership)
takeown /F C:\Folder /R /D Y
icacls C:\Folder /grant %USERNAME%:F /t

:: Check locked files
openfiles /query
```

Useful CMD Batch Scripting

batch

```
@echo off
:: --- Basic error handling ---
set ERRORLEVEL=0

:: --- Check if variable is set ---
if not defined VARNAME (
    echo Variable not set
    set VARNAME=DefaultValue
)

:: --- Loop through files ---
for /f "delims=" %i in ('dir /b *.log') do (
    echo Processing %i
)

:: --- Read file line by line ---
for /f "usebackq delims=" %a in ("file.txt") do (
```

```

        echo %%a
    )

:: --- Pause and wait for key ---
pause
timeout /t 5

:: --- Output to file ---
echo %DATE% %TIME% - Operation completed >> log.txt

:: --- Check exit code ---
if %ERRORLEVEL% NEQ 0 (
    echo Error occurred
    exit /b %ERRORLEVEL%
)

```

CMD Shortcuts and Special Keys

Shortcut	Description
Ctrl + C	Cancel current command
Ctrl + Z	End of file (EoF)
Ctrl + A	Beginning of line
Ctrl + E	End of line
Ctrl + Left/Right	Word navigation
Ctrl + Up/Down	Command history
F7	Command history dialog
F8	Search command history
F9	Execute specific command number
Tab	Auto-complete paths
Ctrl + R	Search command history (PowerShell)
> file.txt	Redirect output
>> file.txt	Append output
2> errors.txt	Redirect errors
	Pipe output
&	Run command in background
&&	Run next if previous succeeds
	Run next if previous fails

Environment Variables in CMD

Variable	Description	Example
%COMPUTERNAME%	Computer name	echo %COMPUTERNAME%
%USERNAME%	Current username	echo %USERNAME%
%USERPROFILE%	User home directory	cd %USERPROFILE%

Variable	Description	Example
%WINDIR%	Windows directory	dir %WINDIR%
%SYSTEMROOT%	System root	echo %SYSTEMROOT%
%TEMP%	Temporary folder	del %TEMP%*.*
%TMP%	Temporary folder (alt)	cd %TMP%
%PATH%	Command search path	echo %PATH%
%PROCESSOR_ARCHITECTURE%	CPU architecture	echo %PROCESSOR_ARCHITECTURE%
%NUMBER_OF_PROCESSORS%	CPU cores	echo %NUMBER_OF_PROCESSORS%
%DATE%	Current date	echo %DATE%
%TIME%	Current time	echo %TIME%
%ERRORLEVEL%	Last command exit code	echo %ERRORLEVEL%

Important File Locations

Location	Purpose
%WINDIR%\System32\drivers\etc\hosts	Hosts file
%WINDIR%\System32\drivers\etc\lmhosts.sam	LMHOSTS sample
%WINDIR%\System32\drivers\etc\networks	Network names
%WINDIR%\System32\drivers\etc\protocol	Protocol numbers
%WINDIR%\System32\drivers\etc\services	Service ports
%WINDIR%\WindowsUpdate.log	Windows Update log
%WINDIR%\setupapi.log	Device installation log
%WINDIR%\INF\setupapi.dev.log	Device setup log
%USERPROFILE%\AppData\Local\Temp	User temp files
%WINDIR%\System32\winevt\Logs	Event log files

Performance Counters in CMD

cmd

:: List all counters

typeperf -q

:: Monitor CPU

typeperf "\Processor(_Total)\% Processor Time" -sc 10

:: Monitor Memory

typeperf "\Memory\Available MBytes" -sc 10

:: Monitor Disk

typeperf "\PhysicalDisk(_Total)\% Disk Time" -sc 10

:: Monitor Network

typeperf "\Network Interface(*)\Bytes Total/sec" -sc 10

:: Multiple counters

typeperf "\Processor(_Total)\% Processor Time" "\Memory\Available MBytes"
"\PhysicalDisk(_Total)\% Disk Time" -sc 10